

Programming Assignment # 2, Linear/Logistic Regression

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Due Date: 15 Oct 2025

1 Objective

In this exercise we will learn how to fit a linear and logistic regression on given data to see how logistic regression can help us in making decisions. You will do **your own implementation** of logistic regression and get the results. Then you will use some library for the implementation of Logistic regression to compare your results.

2 Goal

Predicting academic outcomes and identifying at-risk students can guide interventions that improve retention and performance. In this assignment, you will:

- Model final exam scores with linear regression.
- Classify dropout risk with logistic regression.
- Combat overfitting and multicollinearity using L1 (Lasso) and L2 (Ridge) regularization.

3 Dataset

Use the *Students Performance in Exams* dataset. This dataset is available on <https://www.kaggle.com/datasets/spscientist/students-performance-in-exams/data>. It includes following:

- **Demographics:** gender, race/ethnicity, parental level of education
- **Test scores:** math score, reading score, writing score
- **Additional features:** lunch type, test preparation course

You may extend these features with synthetic features (e.g., average of two scores, interaction terms).

You can also explore the Kaggle for different codes that are available. Using these codes, you can setup your initial working codes.

4 Task Breakdown

You have to perform the following tasks in this assignment:

4.1 Data Pre-processing & EDA

- Handle missing values and outliers.
- Encode categorical variables.
- Visualize correlations and distributions.

4.2 Linear Regression

- Define target: final exam score (e.g., average of math, reading, writing).
- Split data into training and test sets.
- Train a baseline ordinary least squares model.
- Report RMSE and R^2 .

4.3 Logistic Regression

- Create binary target: dropout risk (e.g., score ≥ 60 as “at risk”).
- Train a standard logistic regression model.
- Report accuracy, precision, recall, and ROC-AUC.

4.4 Regularization

- Train Lasso and Ridge versions for both tasks.
- Perform cross-validation to select penalty parameter α .
- Compare performance against baseline.

4.5 Interpretation

- Identify top positive and negative predictors in each model.
- Discuss how regularization altered coefficients.
- Suggest actionable insights for educators.

5 Deliverables

- A Jupyter/ Colab notebook (or equivalent) containing code, figures, and commentary.
- A written report (4–5 pages) covering methodology, results, and educational implications.
- Answer the following questions with evidence of your claims.
 - Which features do you expect to most influence student performance, and why?
 - Compare unregularized vs regularized models (Ridge vs Lasso) and explain why the performance changed.
 - Is accuracy/RMSE alone sufficient? When can it be misleading?
 - Pick any two misclassified cases and suggest why the model might have failed on them.
 - Which features were most influential and what real world insight does it provide ?
 - If given more time or data, how would you impact the model? How does more data effect your models generally ? Try truncating the data and see what happens to your ,models performance and draw conclusions.

Note: All the claims above should have rigorous evidence attached.

WARNING:

- Try not to copy code or report. Anybody caught doing so, will get zero marks.